

Vehicle path planning in various driving situations based on the elastic band theory for highway collision avoidance

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Abstract

This paper presents an emergency path generation method which could be applied in various driving situations, such as on a straight or a curved road, and in the lane-change and lane-departure scenarios which are the most likely from a traffic viewpoint. It is based on the artificial potential approach and the elastic band theory. The assessment of the emergency path is based on the dynamic performances; the yaw rate and the lateral acceleration of the host vehicle are chosen in this paper. In order to make evasion manoeuvres steadier, a guide potential attached at the front of the obstacle vehicle and a guide potential attached at the rear of the obstacle vehicle are built to affect the moving vehicles in such a way that it encourages the host vehicle to change lane appropriately. Meanwhile, a hazard map of the road environment including the borders and the obstacle vehicles is generated. Thus the forces acting on the corresponding nodes of an elastic band should be zero because of the equilibrium condition; the trajectory with a low hazard is calculated via a numerical method. The simulation results show that the approach is acceptable and leads to a successful collision avoidance manoeuvre in various driving situations.

Keywords

Collision avoidance, elastic band, emergency path planning, artificial potential, vehicle's dynamic performance

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Introduction

Active safety systems such as anti-lock braking systems, adaptive cruise control (ACC) systems and forward collision warning (FCW) systems offer possibilities for improving traffic safety by assisting the driver in his or her driving task. Statistics show that over 60,000 people died because of traffic accidents in 2011 in the People's Republic of China.¹ The number of casualties is huge, but active driver assistance systems (ADASs) that support in guiding and stabilizing the vehicle have the potential to reduce significantly the number of road accidents. A European Field Operational Test study² has demonstrated that cars with both ACC and FCW could potentially affect up to 5.7% of the injury accidents on motorways, and potentially 0.6% of accidents involving trucks. Vehicle collision avoidance is an active field for research on safety. For vehicle emergency path planning, a method based on an artificial potential field, originally motivated by work in the field of robotics,^{3,4} has been modified for the problem of automotive guidance and extended by comparison with related methods;^{5,6} an important contribution of the

predictive motion planning approach is the incorporation of moving objects. This method has some advantages compared with geometric path planning, as proposed by Brandt et al.⁷ Brandt et al.^{7,8} and Sattel and Brandt⁹ focused on lateral guidance strategies based on the environment using the Global Positioning System and sensor data. Further research on path planning with a potential field based on elastic bands⁹ gave a more specific and systematic approach. An extension of the elastic bands path-planning method

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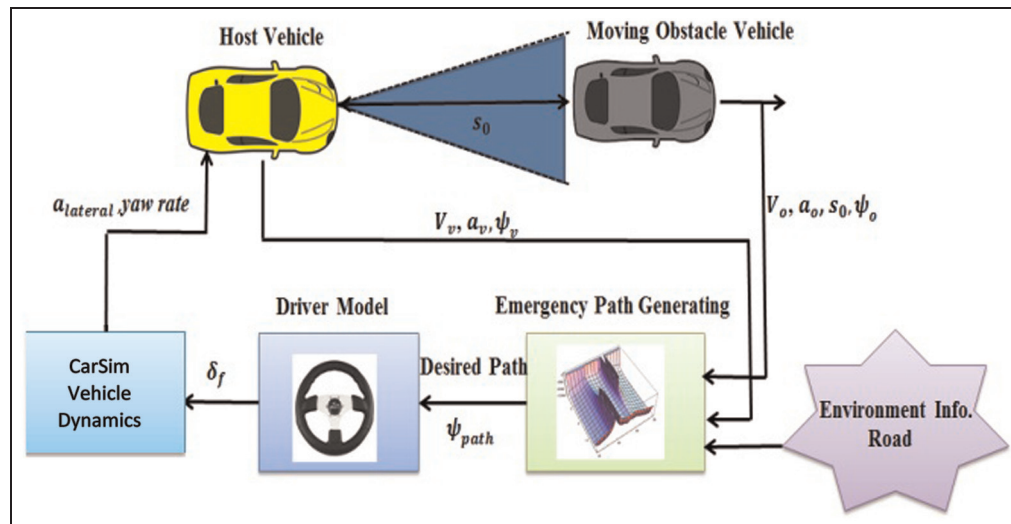


Figure 1. A possible scheme for the whole simulation.
Info: information.

incorporating vehicle dynamics has been proposed by Hesse and Sattel¹⁰ and Hesse et al.¹¹

The artificial potential method has been used widely in vehicle lane-keeping and lane-changing scenarios; the dynamics of the vehicle are coupled with the environment, which ensures that the system exhibits safe motion in the absence of driver inputs.^{12,13} It has also been used for collision avoidance on the highway.¹⁴ There are also some other approaches for path planning in an emergency situation. For example, with a propagation method, possible and unavoidable collisions have been identified to derive evasion trajectories in scenarios with multiple objects.¹⁵ Beyond that, a tree search algorithm and mixed-integer linear programming for multiple objects could be used; this has been described by Frese and Beyereer.¹⁶

According to a survey for the vehicle accidents, the most frequent accident scenarios can be listed as follows:¹⁷

- a collision with another vehicle that is driving in front;
- a collision with another vehicle that is moving laterally while travelling in the same direction;
- a collision with another vehicle that is turning into the road;
- a collision with another vehicle that is travelling in the opposite direction;
- a collision with a static obstacle vehicle on the road.

Therefore, we simulate all these scenarios, generating an emergency path based on an artificial potential and the elastic band theory. It deals with the problem such as making evasion manoeuvres on a curved road, or avoiding an obstacle vehicle by a lane-change or lane-departure manoeuvre. These are rarely applied, and

previous research studies on active safety have utilized the elastic band theory. Moreover, in order to make evasion manoeuvres safer, in comparison with the original elastic band approach, an extra guide potential is attached at the front of the obstacle vehicle and an extra guide potential is attached at the rear of the obstacle vehicle in order to affect the moving vehicles in such a way that the host vehicle is encouraged to change lane appropriately. We have already studied the influencing factors of emergency path planning in our previous research; more specifically, the guide potential shape, the guide potential influencing factor, the stiffness value of the elastic band and the speed of the host vehicle have been studied and analysed by Song et al.¹⁸

Meanwhile, a hazard map of the road environment which includes the borders and the obstacle vehicles is generated. Thus the forces acting on the corresponding nodes of an elastic band should be zero owing to the equilibrium condition. A trajectory with a low hazard is calculated via a numerical method, namely the Newton–Raphson method. We also describe an easy geometric procedure to generate the initial solution. Finally, simulations of the path following are realized using CarSim software to verify whether the planning path is feasible or not. The assessment of the emergency path is based on the dynamic performance of the host vehicle; the yaw rate and the lateral acceleration are chosen in this paper.

A possible scheme for the whole simulation is depicted in the Figure 1; information on the obstacle's velocity, acceleration and orientation is obtained via the vehicle sensors and, in combination with the road information, is used for emergency path planning. Then the simulated driver follows the planned trajectory, cooperating with the dynamic vehicle model; the yaw rate and the lateral acceleration are obtained to judge whether the vehicle is stable or not in the whole evasion manoeuvre.

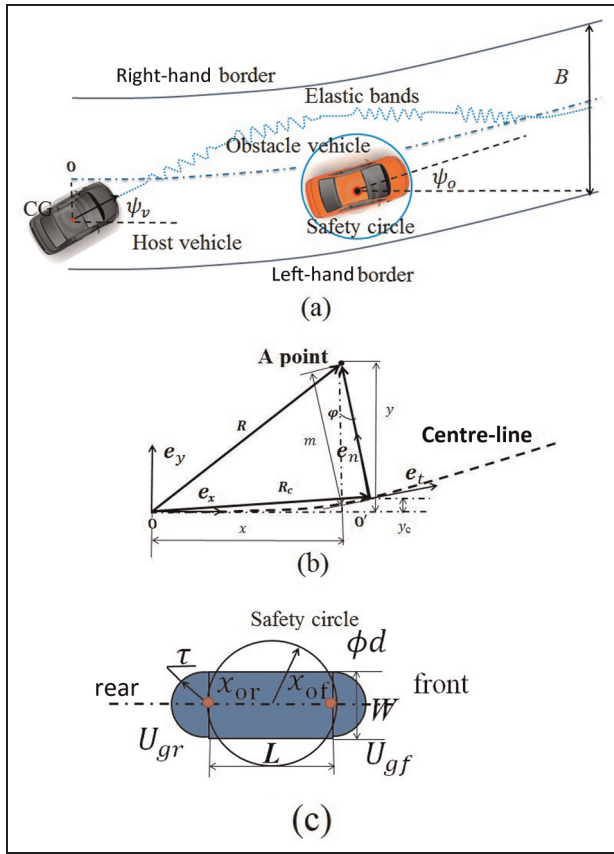


Figure 2. Geometry of the road and guide potential.
CG: centre of gravity.

Road geometry implementation

The environment sensors detected the main course of the road, and the width B of the road is identified as well. As shown in Figure 2(a), the origin O of the road's fixed reference frame is defined as a point on the road's centre-line corresponding to the centre of gravity of the host vehicle at the planning instant. The road's centre-line is constructed from a clothoid curve. A clothoid is a curve whose curvature changes linearly with its curve length. For a clothoid segment, its curvature c starts at c_0 and varies linearly with the arc length l . The curvature at a certain point of the clothoid after length l is given by

$$c(l) = c_0 + c_1 l \quad (1)$$

where the parameters are the initial curvature c_0 , the curvature change rate c_1 and the length l of the segment. Since the clothoid cannot be evaluated in a closed form, clothoid segments are often approximated by a polynomial in the road's fixed frame, which is given by

$$y_c(x) = y_0 + \delta_0 x + \frac{1}{2} c_0 x^2 + \frac{1}{6} c_1 x^3 \quad (2)$$

where y_c represents the value of the y axis of the centre-line in the road's reference coordinate frame and x is the horizontal distance to the origin of the road's fixed

reference frame. The initial lateral offset and heading direction are denoted as y_0 and δ_0 respectively. The polynomial model is sufficiently precise as long as the change in the heading direction δ_0 of the clothoid segment is less than 15° . The position vector R_c of the road's centre-line in Cartesian coordinates, as illustrated in Figure 2(b), is given by

$$R_c = x e_x + y_c e_y \quad (3)$$

Obviously, with the boundary conditions $y_c(0) = 0$ and $y'_c(0) = 0$, the lateral offset y_0 and the tangent of the initial heading direction δ_0 vanish. The unit vectors in the normal–tangential coordinates system along the road's centre-line are

$$e_t = \frac{R'_c}{\|R'_c\|} = \frac{1}{\sqrt{1 + y'^2_c}} e_x + \frac{y'_c}{\sqrt{1 + y'^2_c}} e_y$$

$$e_n = \frac{-y'_c}{\sqrt{1 + y'^2_c}} e_x + \frac{1}{\sqrt{1 + y'^2_c}} e_y \quad (4)$$

Thus, the corresponding points at the borders of the road are given by

$$R_b = R_c \pm \frac{B}{2} e_n \quad (5)$$

where R_b denotes the position vector of the road borders in the road's reference coordinate frame. For any point on the road, the position vector, which is illustrated in Figure 2(b), is given by

$$R = x e_x + y e_y$$

$$= R_c + m e_n \quad (6)$$

where m is a scalar value along the e_n component; if we consider that the angle ϕ of the clothoid segment of the road is small, with a small-curvature assumption we obtain $m \approx y - y_c$, so that the position vector of a point on the road can be also expressed approximately as

$$R \approx R_c + (y - y_c) e_n \quad (6)$$

Internal potential in the elastic band

We assume that the collision-free path is controlled by an elastic band,⁶ as depicted in Figure 2(a). The elastic band is composed of N discrete nodes. Each node P_i of the elastic band is represented by its position vector r_i with respect to the centre of the vehicle's reference frame at the planning instant t_0 . The position vector of each node is denoted by

$$r_i = x_i e_x + y_i e_y \quad (7)$$

The node P_0 is fixed on the vehicle and the end node P_N is also fixed ahead far away from the vehicle for a given planning length. Thus, the boundary condition of the elastic band is given by

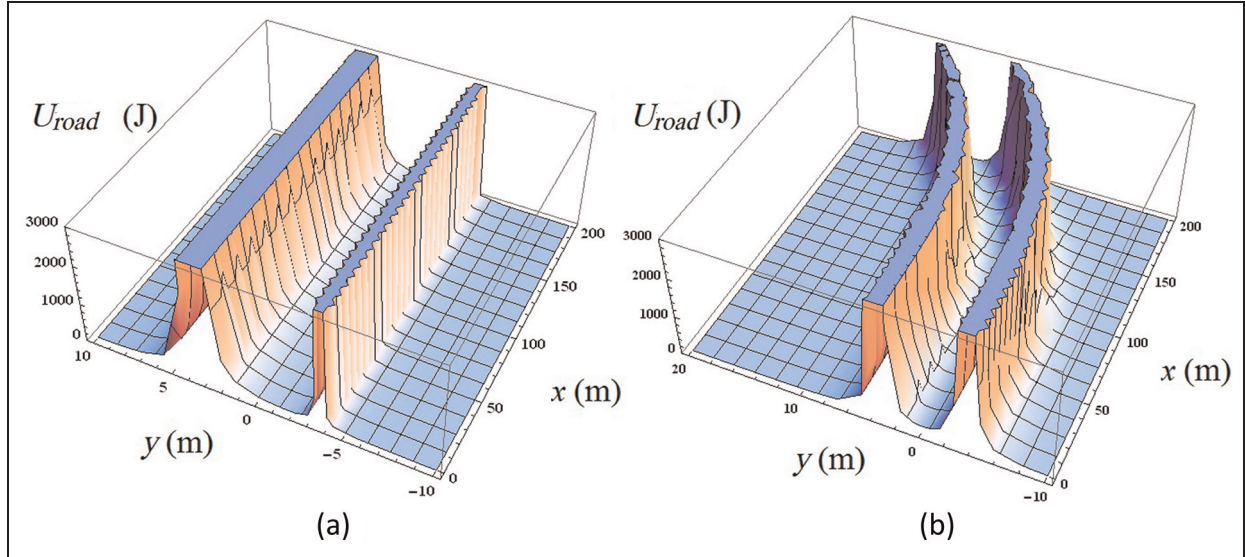


Figure 3. Examples of the road's potential for (a) a straight road and (b) a curved road.

$$\mathbf{r}_0 = \begin{bmatrix} x_0 \\ y_0 \end{bmatrix}, \mathbf{r}_N = \begin{bmatrix} x_N \\ y_N \end{bmatrix}$$

Assuming that the transitional springs are linear, each node is immersed in a quadratic internal potential field and is affected by its neighbouring nodes; the quadratic internal potential field is given by

$$U_{int,i} = \frac{1}{2} [k_{i-1}(\|\mathbf{r}_i - \mathbf{r}_{i-1}\| - l_{i-1})^2 + k_i(\|\mathbf{r}_{i+1} - \mathbf{r}_i\| - l_i)^2] \quad (9)$$

with $i = 1, 2, 3, \dots, N-1$ and where k_i and l_i represent the spring constant and the unstretched length respectively of the spring. The Euclidean norm is defined as

$$\|\mathbf{r}_i - \mathbf{r}_{i-1}\| = \sqrt{(x_i - x_{i-1})^2 + (y_i - y_{i-1})^2} \quad (10)$$

Road potentials

One of the key stages parts is the creation of an appropriate potential function $U(\mathbf{r})$ to constrain the motion and the position of the vehicle. According to the application in the robotic field, the configuration space should have the following characteristics.¹⁴

1. The vehicle should have a preference to stay in well-defined lanes and further to remain in the middle of a lane. Furthermore, the direction of travel is limited on the highway.
2. The lateral velocity due to a lane change is generally small in comparison with the longitudinal velocity. Accordingly, acceptable tolerances for proximity of the obstacle vehicle vary with the direction; i.e. being 1 m to the side of another vehicle is more acceptable than following 1 m behind. Thus, we present a potential function for highway

driving that encodes avoiding road edges while preferring travel in a lane centre; this is described in detail in the following section.

The potential of the road borders prevents the vehicle from leaving the highway. When the vehicle comes close to the edge of the roads, the value of the potential should become infinite. Thus, a repulsive potential is employed according to

$$U_{border,q} = \frac{\eta_q}{2} \left(\frac{1}{\|\mathbf{R} - \mathbf{R}_{b,q}\|} \right)^2 \quad (11)$$

where η is a scaling factor and $q \in [l, r]$, which denotes the position of the left-hand border or the right-hand border. The road's potential is the sum of the potentials of both sides. In a normal situation when there are no obstacles around to be detected, the host vehicle prefers to move along the centre of one lane. Noting that the value of the virtual force derived from the road's potential is calculated from $-\nabla U_{border,q}$, therefore the force of each side should be equal and the ratio of η_l to η_r can be obtained. In particular, when moving on a straight road, the relationship between η_l and η_r should satisfy $\eta_l/\eta_r = (0.75B/0.25B)^3 = 27$. Figure 3 shows an example of the road's potential on a straight road and on a curved road respectively.

Potentials of obstacles

The potential of the obstacle vehicle aims to keep the host vehicle a safe distance away from the obstacle vehicle by building a potential that rises to infinite strength when any other vehicles approach any part of the it. Additionally, the potential must be shaped so that a lane change is encouraged if the host vehicle is rapidly approaching from behind. The vehicle should be forced away rather than stopped. We choose a logarithmic

potential⁴ as the obstacle vehicle's potential, which is given by

$$U_{car} = -k_{car} \ln \left(\| \mathbf{R} - \mathbf{R}_o^{(m)} \| - \frac{d}{2} \right) \quad (12)$$

where $\| \mathbf{R} - \mathbf{R}_o^{(m)} \|$ denotes a distance measurement to the m th obstacle vehicle and k_{car} is the scaling factor of the obstacle vehicle. We also introduce the safety circle as detailed by Quinlan and Khatib;⁴ for simplicity, d is the diameter of the safety circle, as depicted in Figure 2(c).

Besides the obstacle vehicle's potential, in order to guide the vehicle so that a collision with the obstacle vehicle is avoided (especially when the host vehicle is behind the obstacle vehicle), the potential function influence should occur at a larger distance. We focus on the situation of a straight road; thus, an extra guide potential attached at the front and an extra guide potential attached at the rear of the obstacle vehicle are proposed, keeping the host vehicle away from the obstacle vehicle in the corresponding direction. A half-circle with a radius τ is added behind the vehicle. Set $\mathbf{R}_{or} = (x_{or}, y_{or})^T$ as the position vector of the middle point of the obstacle vehicle's left-hand border in the road's reference coordinate frame, as depicted in Figure 2(c). The guide potential at the rear of the obstacle vehicle is denoted by

$$U_{gr}^{(m)} = \begin{cases} -k_r \ln(\| \mathbf{R}^* - \mathbf{R}_{or}^{(m)} \| - \tau), & x \leq x_{or} \\ 0, & \text{otherwise} \end{cases} \quad (13)$$

where $\mathbf{R}^*$ represents the position vector after increasing the influence of the guide potential. Also it is impossible to enter this area. In fact, it acts as an additional safety area behind the obstacle vehicle. In order to shrink the effective distance as proposed above, when calculating $\mathbf{R}^*$, on a straight road, we need to modify only x which is the $\mathbf{e}_x$ component of the position vector $\mathbf{R}$ to x^* ; thus $\mathbf{R}^* = (x^*, y)^T$. We also hope that this effect has a relationship with the host vehicle's relative speed to the obstacle vehicle; therefore, we set

$$\frac{x^* - x_{or}}{x - x_{or}} = e^{-\beta(v-v_m)} \quad (14)$$

Set $\lambda = e^{-\beta(v-v_m)}$, where β is called the influencing factor of the guide potential, v is the speed of the host vehicle, v_m is the speed of the m th obstacle vehicle and λ is the scale term. For any given point on a straight road, we can obtain x^* from equation (14); in fact, it can be regarded as a function with respect to x , namely

$$f: x \rightarrow x_{or} + \lambda(x - x_{or}) \quad (15)$$

This means that a point after it is transformed on the road is actually closer to the obstacle vehicle; therefore, it has a higher potential and can keep the host vehicle away from the obstacle vehicle, which could be much safer. Analogously, a guide potential is also attached at the front of the obstacle vehicle according to

$$U_{gr}^{(m)} = \begin{cases} -k_r \ln(\| \mathbf{R}^* - \mathbf{R}_{or}^{(m)} \| - \tau), & x \leq x_{of} \\ 0, & \text{otherwise} \end{cases} \quad (16)$$

Set $\mathbf{R}_{of} = (x_{of}, y_{of})^T$ as the position vector of the middle point of the obstacle vehicle's right-hand border. It keeps the host vehicle a certain safe distance away when the host vehicle returns to the original lane.

Finally, for M obstacle vehicles, the potential of the obstacle vehicle is given by

$$U_{obst} = \sum_{j=1}^M (U_{car} + U_{gr} + U_{gf})^{(j)} \quad (17)$$

Hazard map

As mentioned above, the hazard map is composed of the road's potential and the potentials of the obstacle vehicles (with M obstacle vehicles) according to

$$U_{ext} = U_{border} + U_{obst} \quad (18)$$

We should pay attention to the fact that the hazard map is not static but varies with the traffic situation at a particular instant in time.

Figure 4, with the three-dimensional plot of the external potential and the corresponding two-dimensional contour plot, shows the difference between an obstacle vehicle with a guide potential attached and an obstacle vehicle without a guide potential. Obviously, with a guide potential, the effect of U_{ext} is increased on a straight road.

Setting the motion trajectory of an obstacle vehicle

In the path-planning procedure, the position of an obstacle vehicle can be predicted via sensor information. However, the motion trajectory of the obstacle vehicle in our simulation is given by a parameterized curve; thus the position of the obstacle vehicle at one moment can be determined. In this paper, we focus on a straight road; thus, two kinds of frequently happening scenarios, namely lane change and lane departure, are considered below.

Lane-change scenario. The trajectory of the lane-change manoeuvre can be represented by an arctangent function or a sinusoid curve; this is depicted in the Figure 5(a) and is described by

$$\Omega := -w \arctan \left(\frac{x - \mu}{\sigma} \right) \quad (19a)$$

where w is the amplitude, μ is the position intersecting with the centre-line of the road, and the value of σ can be used to flatten the curve.

A sinusoid curve can be expressed as

$$\Omega := w \left[\frac{x}{T} - \frac{1}{2\pi} \sin \left(\frac{2\pi}{T} x \right) \right] \quad (19b)$$

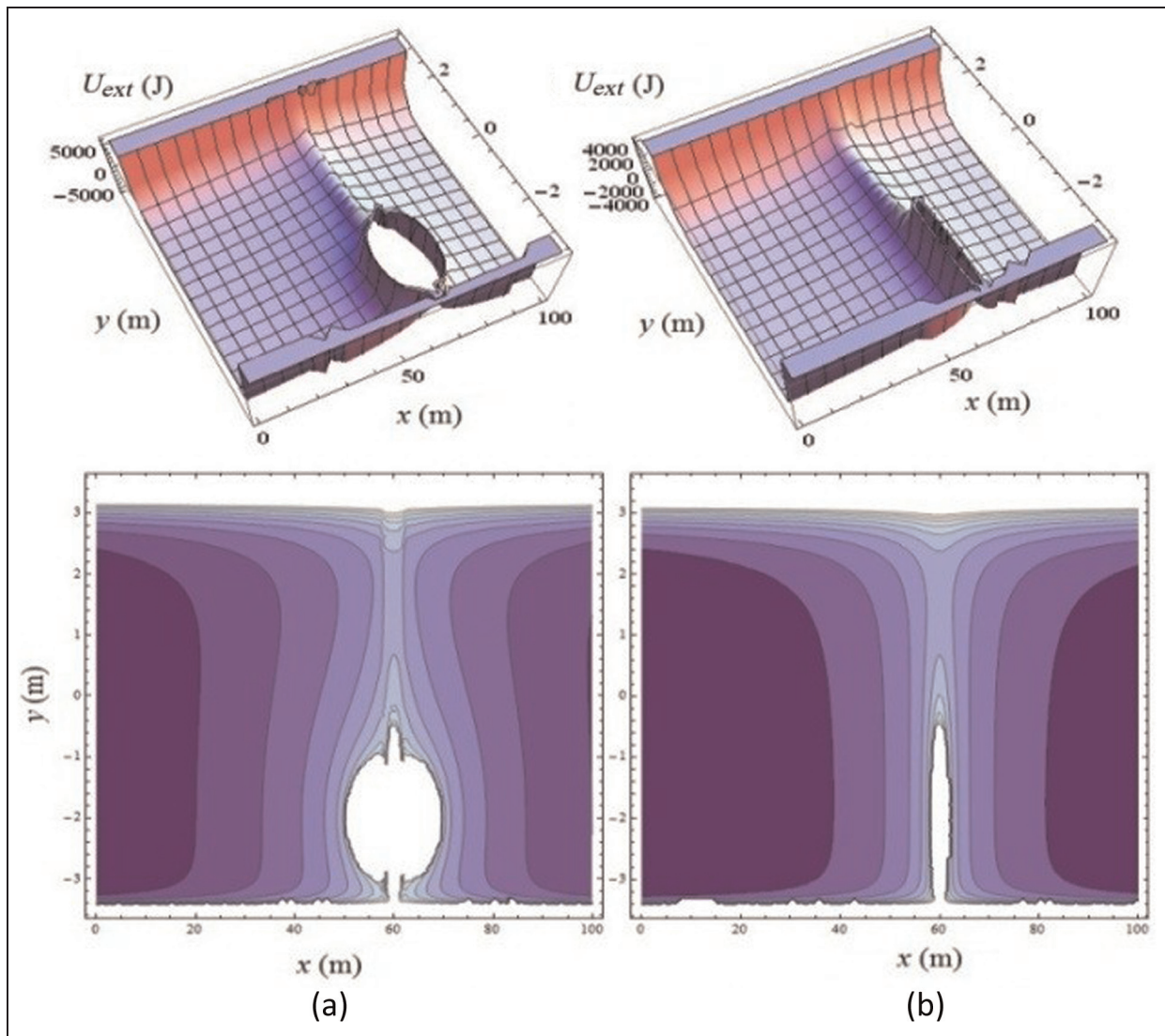


Figure 4. The difference between (a) an obstacle vehicle with a guide potential and (b) an obstacle vehicle without a guide potential.

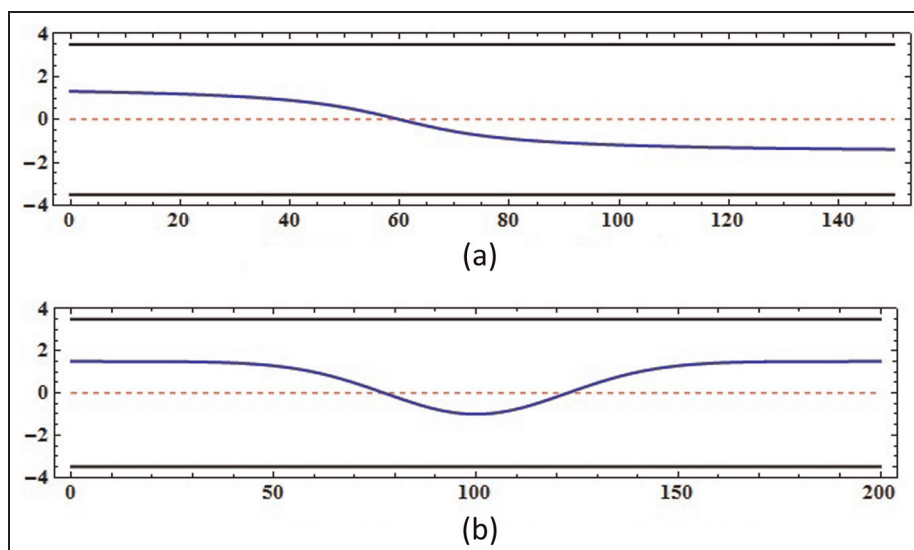


Figure 5. The trajectory of an obstacle vehicle (a) in a lane-change scenario and (b) in a lane-departure scenario.

where w is the amplitude and T determines when the obstacle vehicle finishes to make a lane change. This can also be used for a simulated lane-change trajectory.

Lane-departure scenario. For the lane-departure scenario, the trajectory can be described by a Gauss-like curve; this is shown in Figure 5(b) and is given by

$$\Gamma := a \mp b e^{-[(x-x_{turn})^2/\sigma]} \quad (20)$$

where a represents the initial position of the moving obstacle vehicle, b is the limit position when the obstacle vehicle deviates from its original position, x_{turn} denotes the position when the obstacle vehicle begins to return to its original position and the value of σ can be used to flatten the curve. The scenario can be interpreted as follows: the driver operates the car normally at first and yet the obstacle vehicle leaves its lane because of operation error (careless operation) but the driver did not realize that until he or she drives the obstacle vehicle back to the original lane.

Planning-time calculation. For the host vehicle, the position vector $\mathbf{r}_i$ is evaluated at the planning time t_i when passing one node P_i . Since the equilibrium configuration of the elastic band is determined iteratively, the position vectors $\mathbf{r}_i^{(k)}$ of the previous iteration step are also known. In order to obtain the planning time t_i , the emergency path distance covered when the host vehicle is passing node P_i is needed and is represented by s_i ; it can be approximately evaluated from

$$s_i^{(k+1)} = \sum_{j=1}^i \|\mathbf{r}_j^{(k)} - \mathbf{r}_{j-1}^{(k)}\| \quad (21)$$

where k is the iteration variable, on the assumptions that the host vehicle's initial speed V_v and acceleration a_v are known, that the vehicle starts at $t_0 = 0$ with $s(t_0) = 0$ and that the acceleration a_v remains constant. Thus the speed and the covered distance at one node P_i at the planning time t_i become

$$V_i(t_i) = a_v \Delta t + V_{i-1}(t_{i-1}) \quad (22)$$

$$s_i(t_i) = \frac{a_v}{2} \Delta t^2 + V_{i-1}(t_{i-1}) \Delta t \quad (23)$$

where $V_i(t_i)$ and $s_i(t_i)$ denote the host vehicle's velocity and the covered distance respectively when passing one node P_i at the planning time t_i and where Δt denotes the time interval from t_{i-1} to t_i . Equations (22) and (23) are solved to obtain Δt . Thus the planning time t_i from an algorithmic viewpoint is given by

$$t_i = t_{i-1} + \Delta t$$

$$= \begin{cases} \frac{s_{i,i-1}}{V_{i-1}(t_{i-1})}, & a_v = 0 \\ -\frac{V_{i-1}(t_{i-1})}{a_v} - \sqrt{\left[\frac{V_{i-1}(t_{i-1})}{a_v}\right]^2 + \frac{2}{a_v} s_{i,i-1}}, & a_v < 0 \\ -\frac{V_{i-1}(t_{i-1})}{a_v} + \sqrt{\left[\frac{V_{i-1}(t_{i-1})}{a_v}\right]^2 + \frac{2}{a_v} s_{i,i-1}}, & a_v > 0 \end{cases}$$

$$i = 0, 1, \dots, N \quad (24)$$

with $s_{i,i-1} = s_i - s_{i-1}$ and $V_{i-1}(t_{i-1}) = a_v t_{i-1} + V_v$.

Equilibrium solution computation

Forces acting on the node. The scalar potential field U causes forces to act on each node; to derive these forces, according to the fundamental theorem of vector calculus, the directional derivative is defined as

$$\nabla_i = \begin{bmatrix} \frac{\partial}{\partial x_i} \\ \frac{\partial}{\partial y_i} \end{bmatrix}$$

Thus the forces acting on one node are

$$\begin{aligned} \mathbf{F}_{i,border} &= -\nabla_i U_{border,i} \\ &= \frac{\eta}{(\|\mathbf{R} - \mathbf{R}_b\|)^3} \frac{\mathbf{R} - \mathbf{R}_b}{\|\mathbf{R} - \mathbf{R}_b\|} \end{aligned} \quad (25)$$

$$\begin{aligned} \mathbf{F}_{i,obst} &= -\nabla_i U_{obst,i} \\ &= \sum_{j=1}^M \left[\frac{k_{obst}}{\|\mathbf{R} - \mathbf{R}_o^{(m)}\| - r_{safe}} \frac{\mathbf{R} - \mathbf{R}_o^{(m)}}{\|\mathbf{R} - \mathbf{R}_o^{(m)}\|} \right. \\ &\quad \left. + \frac{k_{g, rear}}{\|\mathbf{R}^* - \mathbf{R}_{or}^{(m)}\| - \tau} \frac{\mathbf{R} - \mathbf{R}_{or}^{(m)}}{\|\mathbf{R} - \mathbf{R}_{or}^{(m)}\|} \right. \\ &\quad \left. + \frac{k_{g, front}}{\|\mathbf{R}^* - \mathbf{R}_{ol}^{(m)}\| - \tau} \frac{\mathbf{R} - \mathbf{R}_{ol}^{(m)}}{\|\mathbf{R} - \mathbf{R}_{ol}^{(m)}\|} \right] \end{aligned} \quad (26)$$

$$\begin{aligned} \mathbf{F}_{i,int} &= -\nabla_i U_{int,i} \\ &= -k_i(\|\mathbf{r}_{i+1} - \mathbf{r}_i\| - l_i) \frac{\mathbf{r}_{i+1} - \mathbf{r}_i}{\|\mathbf{r}_{i+1} - \mathbf{r}_i\|} \\ &\quad - k_{i-1}(\|\mathbf{r}_i - \mathbf{r}_{i-1}\| - l_{i-1}) \frac{\mathbf{r}_i - \mathbf{r}_{i-1}}{\|\mathbf{r}_i - \mathbf{r}_{i-1}\|} \end{aligned} \quad (27)$$

where $\mathbf{F}_{i,border}$, $\mathbf{F}_{i,obst}$ and $\mathbf{F}_{i,int}$ denote the borders' force, the obstacle vehicle's force and the internal force respectively for the node P_i .

Equilibrium conditions. The equilibrium condition for each node means that the sum of the forces acting on the node should be zero according to

$$\mathbf{F}_{i,ext} + \mathbf{F}_{i,int} = 0 \quad (28)$$

with $\mathbf{F}_{i,ext} = \mathbf{F}_{i,road} + \sum_{j=1}^M \mathbf{F}_{i,obst}$. Noting that the starting point and the end point of the elastic band are fixed, we need to solve $2N - 1$ non-linear algebraic equations. The equilibrium configuration of the elastic band is computed at every update step.⁵ Thus, the last equilibrium solution becomes automatically a good starting solution for the next update step. The numerical procedure is described in the next section.

Additionally to the previous section, constraint forces are introduced to allow lateral displacements.⁹ However, for lane-keeping, lane-changing and evasion manoeuvres, longitudinal displacements of the nodes

are not essential, and so the longitudinal position of the node then depends on only the discretization parameter Δx of the elastic band, namely

$$\mathbf{r}_i \cdot \mathbf{e}_x = x_i = i \Delta x, i = 0, \dots, N$$

Thus the constraint forces are eliminated and this results in

$$\mathbf{F}_{i_{\text{ext}},y} + \mathbf{F}_{i_{\text{int}},y} = 0 \quad (29)$$

where $\mathbf{F}_{i_{\text{ext}},y}$ and $\mathbf{F}_{i_{\text{int}},y}$ denote the decomposed external force and the decomposed internal force respectively of one node on the y axis.

Generation of the initial solution. In the paper by Sattel and Brandt,⁹ the initial solution is generated at each planning instant t_0 based on the hazard map and the lateral displacements to the preceding nodes. However, in our paper, an initial solution is generated in a linearly extrapolated way, as shown in Figure 6.

The procedure for generating the initial solution is simple; when the other lane of the road in front is appropriate for a lane-change manoeuvre, calculate the distance between the obstacle vehicle and the border of the road in the $\mathbf{e}_n$ direction. If the vehicle stays in the right-hand (left-hand) lane, then the distance to the left-hand (right-hand) border is calculated. The middle point of the segment is then defined as the initial solution point. The nodes in the area without the obstacle vehicle are then linearly extrapolated between the starting point and those middle points described above, just as illustrated in Figure 6. The initial point P_0 is attached

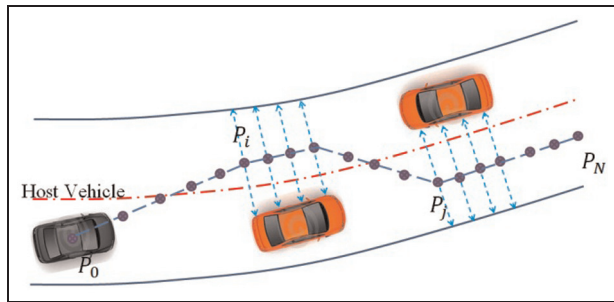


Figure 6. Procedure for generating the initial solution.

to the host vehicle. For a given planning length l_p of the road, we hope that the vehicle can return to its original lane's centre. Thus the position of the goal point P_N is also defined, with $N = \lfloor l_p / \Delta x \rfloor$ (the operator $\lfloor x \rfloor$ here means the maximum integer not larger than x), where Δx represents the horizontal interval length between two neighbouring nodes. Therefore, we define an initial solution in a simple geometric way, as shown in Table 1.

Numerical algorithm for equilibrium equations. This section describes ways to find the equilibrium configuration of the elastic bands. We intend to solve it in a numerical way as the equations are of a non-linear algebraic type. Newton–Raphson iteration is a root-finding algorithm that uses the first few terms of the Taylor series of a function in the vicinity of a suspected root. For numerical solution the problem can be stated in the form

$$\mathbf{F}(\mathbf{z}) = 0, \dim \mathbf{z} = N + 1 \quad (30)$$

where $\mathbf{F} = (\mathbf{F}_{0_{\text{total}}}, \mathbf{F}_{1_{\text{total}}}, \dots, \mathbf{F}_{N_{\text{total}}})^T$ and $\mathbf{F}_{i_{\text{total}}} = \mathbf{F}_{i_{\text{ext}}} + \mathbf{F}_{i_{\text{int}}}$ is composed of external and internal forces acting on the node. According to the previous section, we find a group of initial solution for the equations, which are denoted as $\mathbf{z}^{(0)} = (\mathbf{z}_0^{(0)}, \mathbf{z}_1^{(0)}, \dots, \mathbf{z}_N^{(0)})^T$.

The Newton–Raphson iteration can be interpreted as the linear approximation

$$\mathbf{F}(\mathbf{z}) \approx \mathbf{F}(\mathbf{z}^{(k)}) + \mathbf{J}(\mathbf{z}^{(k)}) (\mathbf{z} - \mathbf{z}^{(k)}) \quad (31)$$

where the matrix $\mathbf{J}$ denotes the Jacobian of $\mathbf{F}$. Equation (31) is obtained by keeping the Taylor series expansion of $\mathbf{F}$ at $\mathbf{z}^{(k)}$ only to first order. Then we need to solve only a linear equation $\mathbf{F}(\mathbf{z}) = 0$, setting the solution as $\mathbf{z}^{(k+1)}$. Performing an iteration step yields

$$\mathbf{z}^{(k+1)} = \mathbf{z}^{(k)} - \mathbf{J}^{-1}(\mathbf{z}^{(k)}) \mathbf{F}(\mathbf{z}^{(k)}), k = 0, 1, 2, \dots \quad (32)$$

Starting from an initial vector $\mathbf{z}^{(0)}$ which is sufficiently close to the solution, the iteration converges quadratically to the root of the equations.

The pseudo-code for the Newton–Raphson algorithm is given in Table 2.

Theoretically, the algorithm could deal with multiple obstacle vehicles; however, with an increasing number

Table 1. Initial solution algorithm.

Algorithm

The initial solution can be generated as follows.

1. For M obstacle vehicles, set $P_m[0] := P[0]$ and $P_m[2M + 1] := P[N]$.

2. Find the first middle point $P_m[2k - 1]$ of the segment between the obstacle vehicle and the road edge in the $\mathbf{e}_n$ orientation with the k th obstacle vehicle, and the same with the second middle point $P_m[2k]$.

If the obstacle vehicle stays in the right-hand lane, then the segment length is the distance to the left-hand border;

Else calculate the distance to the right-hand border.

3. Calculate the linear equation between those points P_m .

4. All the elastic band nodes $P[i]$ are linearly extrapolated with the linear equation calculated above. Thus an initial solution is generated.

Table 2. Newton–Raphson algorithm.**Algorithm**

The Newton–Raphson iteration method is as follows.

1. Set $\mathbf{z}^{(0)}$, $k = 0$.
2. When $|\Delta \mathbf{z}^{(k)}| > \varepsilon$,
 $\Delta \mathbf{z}^{(k)} = -\mathbf{J}^{-1}(\mathbf{z}^{(k)})\mathbf{F}(\mathbf{z}^{(k)})$,
 $\mathbf{z}^{(k+1)} = \mathbf{z}^{(k)} + \Delta \mathbf{z}^{(k)}$,
 $k = k + 1$.
3. End.

Table 3. Parameters of the simulation.

Parameter	Value	Units	Description
B	7	m	Road width
η	100	—	Scale of the borders
l_0	1.2	m	Relaxed spring length
k_{car}	1000	N/m	Stiffness of the potential of the obstacle vehicle
k_{guide}	1000	N/m	Stiffness of the guide potential
k	400,000	N/m	Stiffness of the elastic bands
ε	0.001	m	Tolerance of the Newton–Raphson iteration method
W	2	m	Width of the obstacle vehicle
L	4	m	Length of the obstacle vehicle
d	4	m	Diameter of the safety circle
τ	1	m	Radius of the guide potential

of obstacle vehicles, the costs of finding the solution will become huge, and next we consider only a low-hazard situation. However, at the beginning of the planning procedure, information on the obstacle vehicles (the initial position, the velocity, the moving trajectory, etc.) should be acquired accurately. If there are nodes in the safety circle region of the obstacle vehicles, the path is unfeasible. Therefore the host vehicle will slow down to maintain a safe distance from the obstacle vehicles ahead.

Simulation

In this section, six scenarios which are the most common traffic incident scenes are simulated. The general parameters in the whole simulation are shown in Table 3. Once the path generation is completed, the scenarios are simulated in CarSim as well. Thus, the yaw rate of the host vehicle as an index of the dynamic performance can be obtained when following the emergency path. At the planning instant t_0 , the position, the velocity and the orientation of the obstacle vehicle are obtained from the sensor; with these data the motion of the obstacle vehicle can be predicted within the planning time. Additionally, the planning path is linearly extrapolated. As the traffic scenarios may undergo temporal changes, the hazard map is updated at the time that the host vehicle passes every two nodes. Therefore, in each update of the hazard map, the positions, the velocities

and the orientations are also updated. According to the Road Traffic Safety Law in the People's Republic of China, the maximum speed of a vehicle on the highway is no more than 120 km/h but not less than 50 km/h; thus the speed of the host vehicle in this paper is set at the limit speed of 120 km/h. If the emergency path in a case when the host vehicle moves at the limit speed is acceptable, then it can be considered that it is also feasible for the emergency path when the vehicle moves at a lower speed.

In the simulation results in each of the following figures (Figure 7 to Figure 11), the host vehicle is represented in black, the obstacle vehicle is represented in red, the travelled planned path is a dashed black curve and the future planned path is a solid black curve. The obstacle vehicles are represented by black ellipses and, if they do not move in a straight line, the trajectories of the obstacle vehicles are black dot-dashed curves. Moreover, when the host vehicle reaches the goal point and makes an evasion manoeuvre successively, the actual followed path of the host vehicle is obtained from CarSim and is represented by a solid curve; the target planned path is a red dashed curve.

Case I: static obstacle vehicle. First, we start from the easiest scenario which deals with static obstacle vehicles. A vehicle may come across static obstacle vehicles ahead when driving along the highway. A scenario setting is interpreted as follows: at the beginning of the planning instant, one obstacle vehicle stays in the right-hand lane 50 m ahead of the host vehicle on a straight highway; the coordinates in the road's reference frame are (50, −2.5). Another obstacle vehicle stays in the left-hand lane 150 m away; its coordinates in the road's reference frame are (150, 0). The speed of the host vehicle is 120 km/h, as depicted in Figure 7(a). The simulation results are shown in Figure 7(b). It can be clearly seen that the vehicle successfully avoids clashes with the obstacle vehicles. The actual path followed and the target emergency path are shown in Figure 7(c); the following results are rather satisfying and acceptable.

The yaw rate and the lateral acceleration of the host vehicle when following the emergency path are depicted in Figure 7(d) and Figure 7(e) respectively, and the maximum yaw rate and the maximum lateral acceleration in the manoeuvre are around 11 deg/s and 0.6 g m/s² respectively. The host vehicle still remains stable in our simulation; thus the trajectory is considered acceptable.

Case II: obstacle vehicle that is turning into the road. The host vehicle moves at a constant speed of 25 m/s (90 km/h) on a straight highway. An obstacle vehicle suddenly turns in front of the host vehicle and into the same lane as the host vehicle from a shoulder at a very low initial speed of 5 m/s and then accelerates at rather a high rate of 4 m/s² which remains constant until it finally reaches its maximum speed.

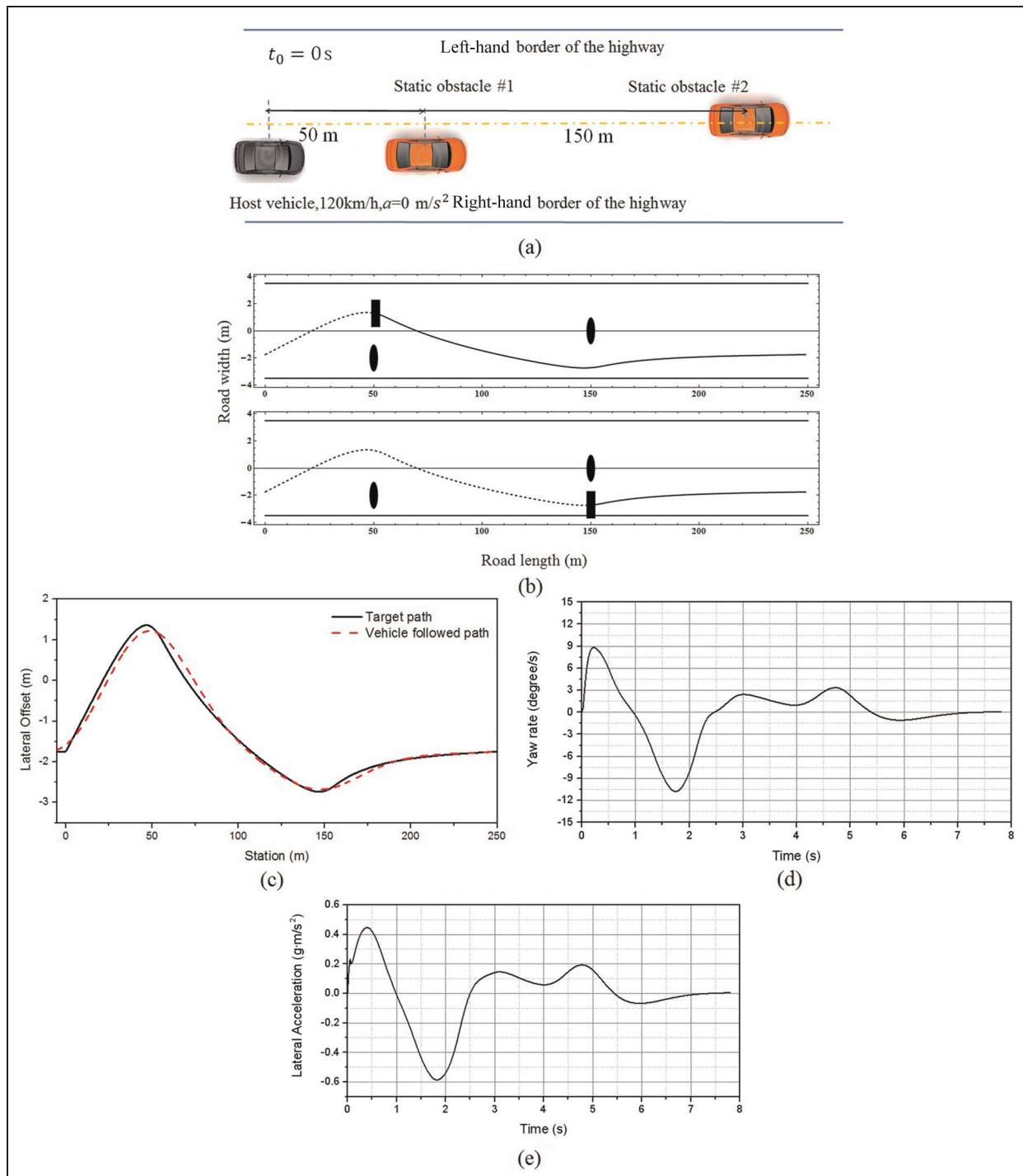


Figure 7. Case I: static obstacle vehicles: (a) scenario; (b) simulation results; (c) the actual path followed and the target emergency path; (d) the maximum yaw rate of the host vehicle; (e) the lateral acceleration of the host vehicle.

The initial distance after the obstacle vehicle turns into the lane is 40 m away from the host vehicle, which is depicted in Figure 8(a). If the host vehicle does not intend to make an emergency manoeuvre, at a speed of 25 m/s and moving straight ahead, the two vehicles will crash into each other after around 2.75 s. Thus, the necessary evasive action should be taken.

For the scenario that we are considering, the obstacle vehicle moves at a low speed initially;

however, it has rather a high acceleration, and so there will be two encounters between the obstacle vehicle and the host vehicle. The obstacle vehicle will move increasingly faster after their first encounter, and the obstacle vehicle will pass the host vehicle eventually. Thus, in the process of path planning, the host vehicle returns to its original lane until the obstacle vehicle passes it at a certain safe distance away. The simulation result is depicted in Figure

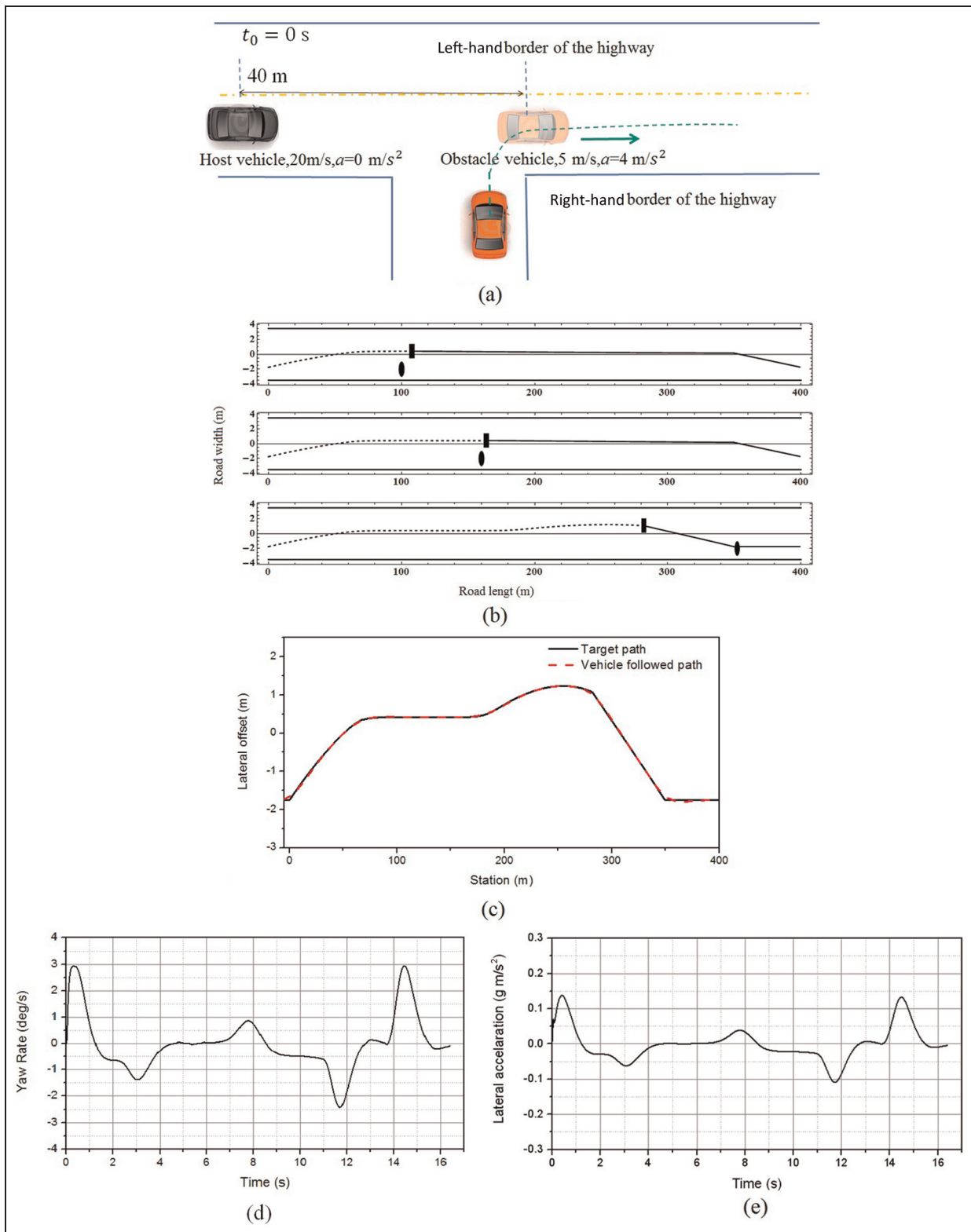


Figure 8. Case II: obstacle vehicle that is turning into the road: (a) scenario; (b) simulation results; (c) the actual path followed and the target emergency path; (d) the maximum yaw rate of the host vehicle; (e) the lateral acceleration of the host vehicle.

8(b). Figure 8(c) shows that the actual followed path of the host vehicle matches the desired planned path quite well. The maximum yaw rate in the manoeuvre

is about 3 deg/s when following the trajectory shown in Figure 8(d). Therefore, the given trajectory is considered to be feasible.

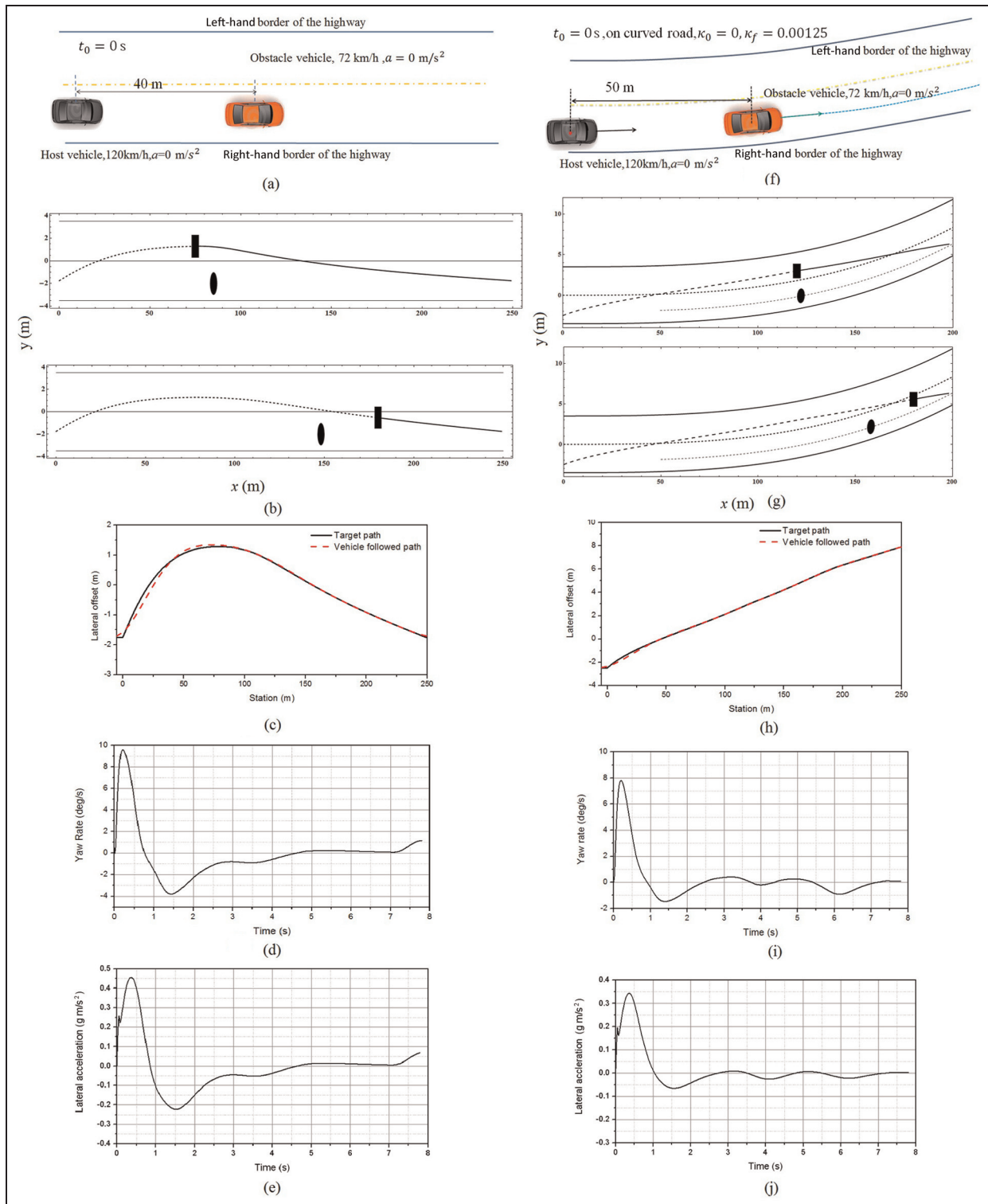


Figure 9. Case III: obstacle vehicle that is driving in front (a)–(e) on a straight road and (f)–(j) on a curved road, (a), (f) scenarios; (b), (g) simulation results; (c), (h) the actual paths followed and the target emergency paths; (d), (i) the maximum yaw rates of the host vehicle; (e), (j) the lateral accelerations of the host vehicle.

Case III: obstacle vehicle that is driving in front. Two kinds of road are considered in this situation.

1. *On a straight road.* We consider a scenario which is depicted in Figure 9(a), namely the host vehicle moves at a constant speed (120 km/h) in the middle

of the right-hand lane alone on the highway, and the obstacle vehicle travels in the same lane at a lower speed (72 km/h) in front of the host vehicle 40 m away. The results are shown in Figure 9(b); the vehicle successfully avoids crashes with the obstacle vehicle. The actual followed path and the

target path are shown in Figure 9(c). The maximum yaw rate and the maximum lateral acceleration in the manoeuvre are about 3 deg/s and 0.45 g m/s² respectively, as shown in Figure 9(d) and Figure 9(e) correspondingly. Therefore, the given trajectory is considered to be feasible.

2. *On a curved road.* The emergency path planning can also deal with a curved road. We construct a curved road using the method described in the implementation of the road geometry. The initial curvature of the road is 0; the final curvature of road is 0.001 25 within a horizontal distance of 200 m. The initial coordinates of the obstacle vehicle moving at a constant speed of 72 km/h (20 m/s) along the road are (50, -2). The constant speed of the host vehicle is 120 km/h (around 33.3 m/s), as depicted in Figure 9(f). The host vehicle will stay in the right-hand lane for a longer distance. The emergency path planning results are shown in the Figure 9(g); the host vehicle successfully avoids crashes with the obstacle vehicle ahead on the curved road. The actual followed path and the target path are shown in Figure 9(h). The maximum yaw rate and the maximum lateral acceleration in the manoeuvre are about 7.8 deg/s and 0.34 g m/s² respectively, as shown in Figure 9(i) and Figure 9(j) correspondingly. Therefore, the given trajectory is also considered to be feasible.

Case IV: obstacle vehicle with a lane departure while travelling in the opposite direction. Now we consider another scenario depicted in Figure 10(a) which also frequently happens in traffic accidents. The host vehicle is travelling along a one-way traffic road at the limit speed on the highway of 120 km/h (around 33.3 m/s), while an obstacle vehicle is travelling in the opposite direction at a constant speed of 72 km/h (20 m/s) in the other lane. The obstacle vehicle departs from its lane and corrects this soon. The host vehicle detects the departure and makes an evasion manoeuvre. The expression for the trajectory of the obstacle vehicle used here is

$$y(x) = 1.5 - 2\exp\left[-\frac{(x-70)^2}{1000}\right]$$

Figure 10(b) shows the situation when these two vehicles encounter each other, and the host vehicle successfully deals with the emergency. The actual followed path and the target path are shown in Figure 10(c); the results are quite good. The maximum yaw rate and the maximum lateral acceleration in the manoeuvre are around 2.5 deg/s and 0.15 g m/s² respectively, when following the desired trajectory, as shown in Figure 10(d) and Figure 10(e) correspondingly. Therefore, the given trajectory is also considered to be feasible.

Case V: obstacle vehicle with a lane change while travelling in the same direction. As shown in Figure 11(a), at the

planning instant, an obstacle vehicle is moving in the right-hand lane at a constant speed of 72 km/h (20 m/s) in front of the host vehicle 40 m ahead and intends to perform a lane-change manoeuvre. The velocity of the host vehicle is 120 km/h (around 33.3 m/s). The expression for the lane-change manoeuvre of the obstacle vehicle used here is $y(x) = -\arctan[(x-100)/16]$. Figure 11(b) shows the emergency path in several time steps around the instant when these two vehicles meet each other.

It can be clearly seen that the host vehicle avoids a collision with the obstacle vehicle which is making a lane change. The actual following path and the target path are shown in Figure 11(c); at a high vehicle speed, the error in the target path is rather large for the first few seconds but the whole process matches well and is acceptable. The maximum yaw rate and the maximum lateral acceleration in the whole process are about 6 deg/s and 0.28 g m/s² respectively, when following the trajectory shown in Figure 11(d) and Figure 11(e) correspondingly. Therefore, with the limit speed on a highway, the given trajectory is considered to be feasible as well.

Conclusions and future work

This paper presents the emergency path planning in different road and traffic scenarios, including static and moving obstacle vehicles, and forward collision avoidance on a straight road and on a curved road; lane-change and lane-departure scenarios on a straight road are also considered. An extra guide potential is attached at the front of the obstacle vehicle and an extra guide potential is attached at the rear of the obstacle vehicle in order to affect the moving vehicles in such a way that the host vehicle is encouraged to change lane appropriately. The simulation results show that the emergency path planning based on an artificial potential and the elastic band theory is successful and feasible. With a limit speed on the highway, the host vehicle can follow the target planned path well. The yaw rate and the lateral acceleration of the host vehicle are obtained from CarSim, and the results indicate that the host vehicle following the trajectory is stable. This research provides a good path-planning realization method for the development of a collision avoidance system or ADAS for autonomous (or semi-autonomous) vehicles.

Our future research studies include two aspects.

1. A hardware-in-the-loop simulation validation, combined with the throttle and brake controls input of the host vehicle, will be used to realize optimally whole algorithms, and it will deal with more complex traffic situations and multiple obstacle vehicles.
2. An empirical formula relating parameters such as the stiffness of the elastic band and the velocity of the host vehicle will be proposed, because setting most of the parameters is based on the simulation experience in this paper.

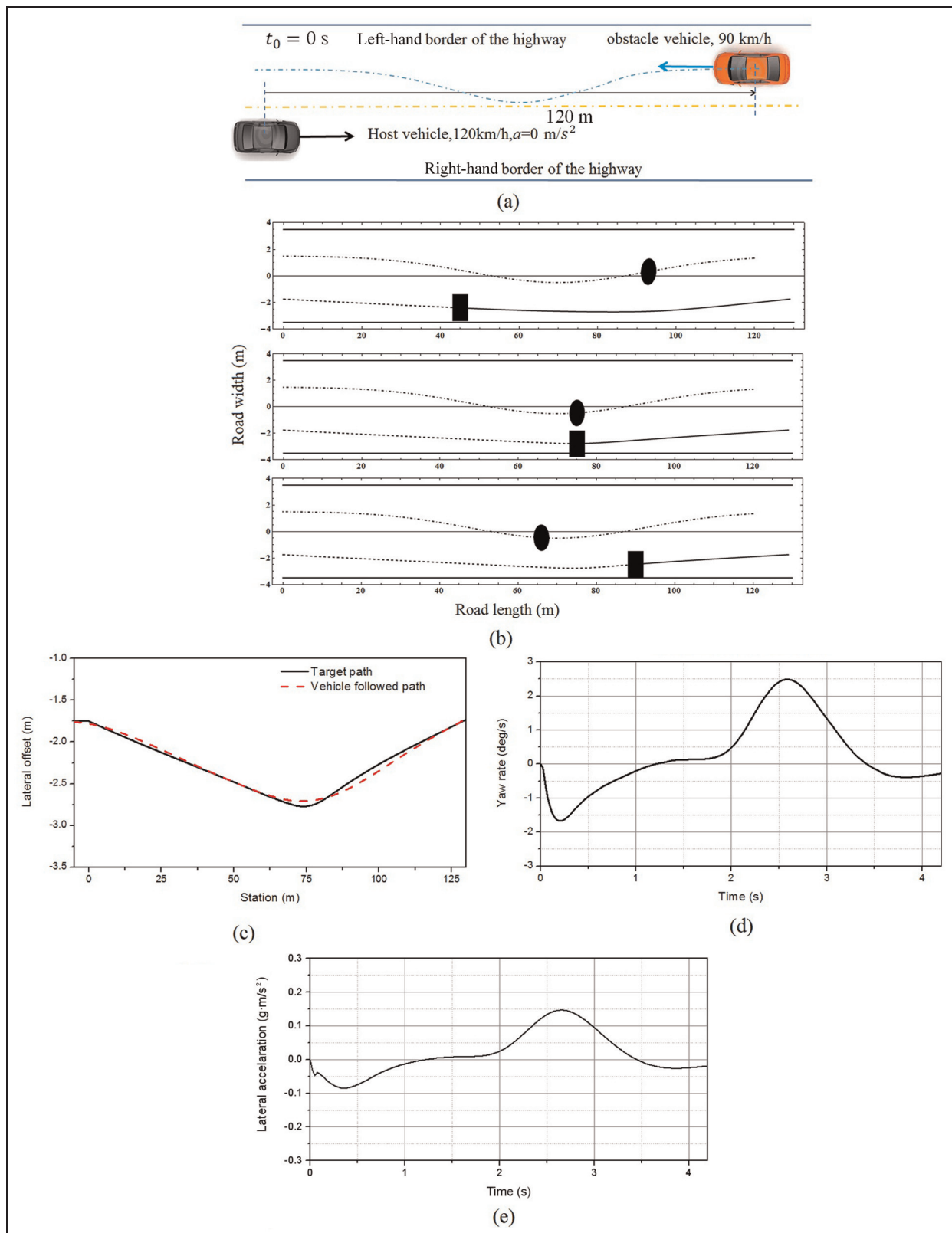


Figure 10. Case IV: obstacle vehicle with a lane departure while travelling in the opposite direction: (a) scenario; (b) simulation results; (c) the actual path followed and the target emergency path; (d) the maximum yaw rate of the host vehicle; (e) the lateral acceleration of the host vehicle.

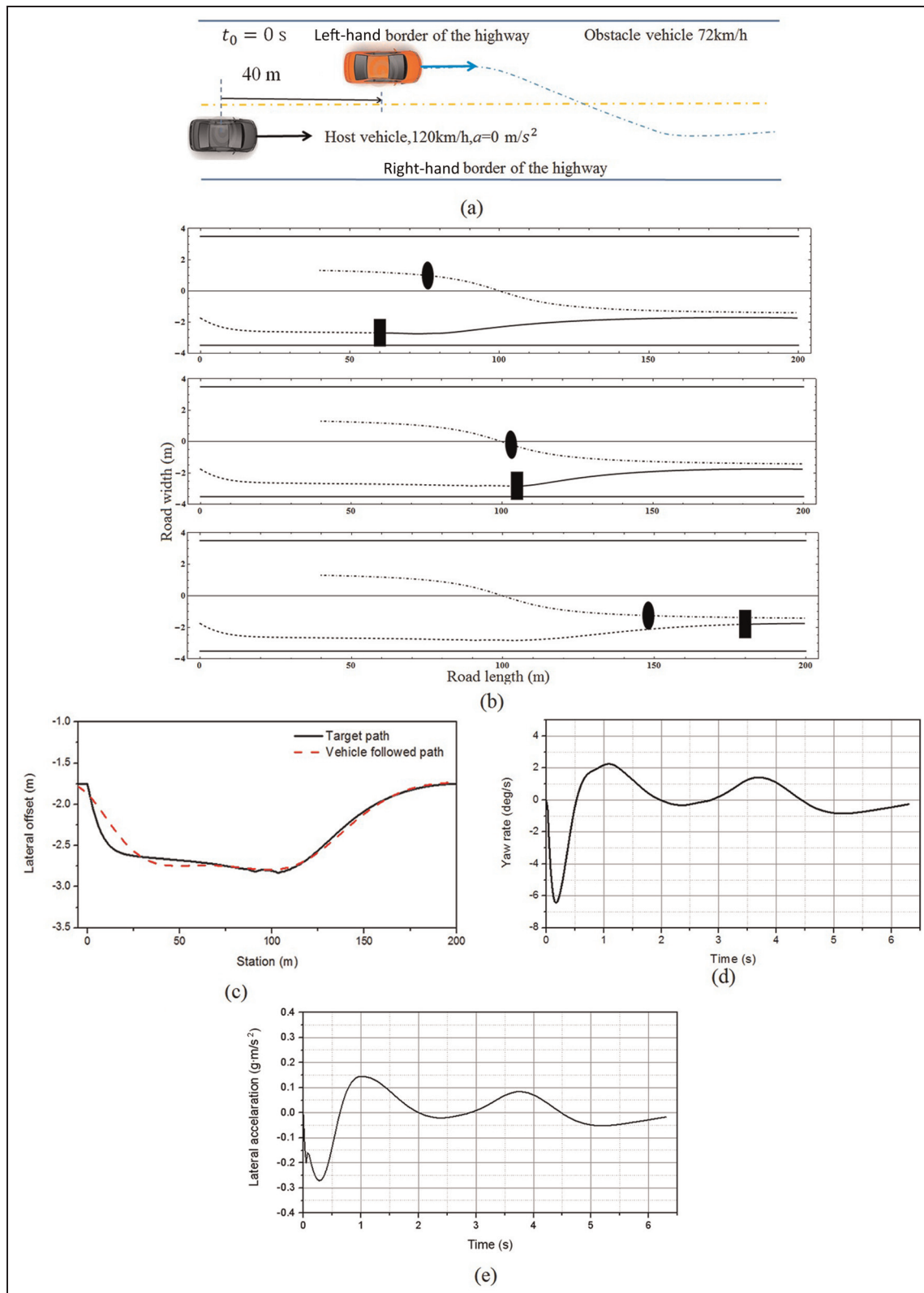


Figure 11. Case V: obstacle vehicle with a lane change while travelling in the same direction: (a) scenario; (b) simulation results; (c) the actual path followed and the target emergency path; (d) the maximum yaw rate of the host vehicle; (e) the lateral acceleration of the host vehicle.

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Appendix I

Notation

a	initial position of the moving obstacle vehicle in a lane departure
a_v	acceleration value of the obstacle vehicle
b	limit position when the obstacle vehicle deviates from its original position
B	road width
c_0	initial curvature of the clothoid segment road
c_1	curvature change rate
d	diameter of the safety circle
e_n	normal unit vector in the normal–tangential coordinates system along the centre-line
e_t	tangent unit vector in the normal–tangential coordinates system along the centre-line
e_x	unit vector of the x axis in the road’s reference coordinates system
e_y	unit vector of the y axis in the road’s reference coordinates system
F_{i_border}	forces caused by the potential of the road borders
F_{i_ext}	external forces acting on the i th node
$F_{i_ext,y}$	external forces acting on the i th node decomposed on the y axis
F_{i_int}	internal forces acting on the i th node
$F_{i_int,y}$	internal forces acting on the i th node decomposed on the y axis
F_{i_obst}	forces caused by the potential of the obstacle vehicle
F_{i_total}	total forces acting on the i th node
J	Jacobian matrix
k	iteration variable
k_{car}	stiffness of the potential of the obstacle vehicle
k_f	stiffness of the front guide potential of the obstacle vehicle
k_i	stiffness of the elastic bands with index i
k_r	stiffness of the rear guide potential of the obstacle vehicle
l	length of the road
l_i	unstretched length of one elastic band
l_p	planning length of the road
L	length of the obstacle vehicle

M	number of obstacle vehicles	W	width of the obstacle vehicle
P_i	elastic band nodes	x	x -axis value in the road's coordinate reference frame
r_i	position vector of the elastic band node in the host vehicle	x_i	x -axis value of the elastic band node in the road's reference coordinate system
R	position vector in the road's reference coordinate system	x_{of}	x -axis value of the middle point of the front side of the obstacle vehicle
R_b	position vector on the road borders in the road's reference coordinate system	x_{or}	x -axis value of the middle point of the rear side of the obstacle vehicle
R_c	position vector on the centre-line in the road's reference coordinate system	x_{turn}	position when the obstacle vehicle begins to return to its original position in a lane departure
R_o	position vector of the obstacle vehicle's centre	y	vertical axis value in the road's reference coordinate system
R_{ol}	position vector of the middle point of the obstacle vehicle's left-hand border	y_0	initial lateral offset of the host vehicle
$R_o^{(m)}$	position vector of the obstacle vehicle in the road's reference coordinate system	y_c	vertical coordinate value of the centre-line in the road's reference coordinate system
R_{or}	position vector of the middle point of the obstacle vehicle's right-hand border in the road's reference coordinate system	y_i	y -axis value of the elastic band node in the road's reference coordinate system
s_i	emergency path distance covered when the host vehicle is passing node P_i	$z^{(0)}$	initial solution of the Newton–Raphson method
$s_{i,i-1}$	distance travelled from the planning time t_i to t_{i-1}	β	influencing factor of the guide potential
t_i	planning time when the host vehicle is passing node P_i	δ_0	initial heading direction of the road's centre-line
U_{border}	potential of the road borders	ε	tolerance of the Newton–Raphson iteration method
U_{car}	potential of the obstacle vehicle itself	η	scaling factor of the potential of the road borders
U_{ext}	whole potential of the external environment	λ	scale term in the guide potential
U_{gr}	rear guide potential of the obstacle vehicle	μ	intersecting point of the lane-change tangent curve with the centre-line of the road
U_{gf}	front guide potential of the obstacle vehicle	σ	parameter flattening the curve in a lane change or lane departure
$U_{int,i}$	internal potential of the elastic bands with index i	τ	radius of the guide potential
U_{obst}	whole potential of the obstacle vehicle	φ	angle of the clothoid segment of the road
v	speed of the host vehicle		
v_m	speed of the m th obstacle vehicle		
V_v	initial velocity value of the host vehicle		
w	amplitude of the lane-change curve		